

Mini Project

Task 1

```
/*
    Create a console-based app called Music Listening
    Logger that allows users to enter the number of minutes
    They listened to music each day
    for a week and stores this data in an array.
*/

#include<stdio.h>

int main()
{
    printf("\n>->->Welcome to Music Listening App<-<-<\n\n");

    int min[100];
    int i;
    printf("Enter the number of minutes you listen to music each days\n\n");
    for(i=1;i<=7;i++)
    {
        printf("Day %d:-",i);
        scanf("%d",&min[i]);
    }

    for(i=1;i<=7;i++)
    {
        printf("Your day %d music play time:- %d Minutes\n",i,min[i]);
    }
}
```

Output:

```
E:\06Aug_Kevin_SE\C_Assign  X  +  v

>->->Welcome to Music Listening App<-<-<

Enter the number of minutes you listen to music each days

Day 1:-23
Day 2:-45
Day 3:-56
Day 4:-23
Day 5:-12
Day 6:-22
Day 7:-10
Your day 1 music play time:- 23 Minutes
Your day 2 music play time:- 45 Minutes
Your day 3 music play time:- 56 Minutes
Your day 4 music play time:- 23 Minutes
Your day 5 music play time:- 12 Minutes
Your day 6 music play time:- 22 Minutes
Your day 7 music play time:- 10 Minutes

-----
Process exited after 25.77 seconds with return value 0
Press any key to continue . . .
```

Task 2

```
/*
    Add a menu-driven interface to your Music Listening
    Logger so users can choose to
    log new listening minutes,
    view their weekly summary,
    or exit the app.
    Hint: Use a loop to repeatedly show the menu until the user selects exit.
*/
```

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int min[100];
```

```
    int i,choice;
```

```
    printf("-----\n");
```

```
    printf("Welcome to Music Listening App\n");
```

```
    printf("-----\n");
```

```
    while(1)
```

```
    {
```

```
        printf("Press 1 to Add your Loggs\n");
```

```
        printf("Press 2 to view Your Loggs\n");
```

```
        printf("Press 3 to Log out\n");
```

```
        printf("-----");
```

```
        printf("\n");
```

```
        scanf("%d",&choice);
```

```
        printf("-----\n");
```

```
        switch(choice)
```

```
        {
```

```
            case 1:
```

```
                for(i=1;i<=7;i++)
```

```
                {
```

```
                    printf("Day %d:-",i);
```

```
                    scanf("%d",&min[i]);
```

```
                }
```

```
                printf("-----\n");
```

```
                break ;
```

```
            case 2:
```

```
                printf("-----\n");
```

```

        printf("Your Added Log\n");
        printf("-----\n");
    for(i=1;i<=7;i++)
    {
        printf("Your day %d music play time:- %d Minutes\n",i,min[i]);
        printf("\n");
    }
        printf("-----\n");
    break;

    case 3:
        printf("-----");
        printf("\nLogging out! Visit Again\n");
        printf("-----");
        printf("\n");
    break;

    default:
        printf("-----");
        printf("\nError!!!! Please go with 1 or 2 or 3 only\n");
        printf("-----");
        printf("\n");
    break;
}

    if(choice==3)
    {
        break;
    }
}

```

Output:

```
E:\06Aug_Kevin_SE\C_Assign  X  +  v

-----
Welcome to Music Listening App
-----
Press 1 to Add your Loggs
Press 2 to view Your Loggs
Press 3 to Log out
-----
1
-----
Day 1:-10
Day 2:-20
Day 3:-
30
Day 4:-40
Day 5:-50
Day 6:-60
Day 7:-52
-----
Press 1 to Add your Loggs
Press 2 to view Your Loggs
Press 3 to Log out
-----
2
-----
Your Added Log
-----
Your day 1 music play time:- 10 Minutes
Your day 2 music play time:- 20 Minutes
Your day 3 music play time:- 30 Minutes
Your day 4 music play time:- 40 Minutes
Your day 5 music play time:- 50 Minutes
Your day 6 music play time:- 60 Minutes
Your day 7 music play time:- 52 Minutes
-----
Press 1 to Add your Loggs
Press 2 to view Your Loggs
Press 3 to Log out
-----
3
-----
Logging out! Visit Again
-----

-----
Process exited after 31.17 seconds with return value 0
Press any key to continue . . . |
```

/* Task 3

Implement file handling in your Music Listening Logger
so that when the user logs their daily listening minutes,
the data is saved to a file
named music_log.txt for persistence.

*/

#include<stdio.h>

int main()

{

int min[100];

int i,choice;

printf("-----\n");

printf("Welcome to Music Listening App\n");

printf("-----\n");

while(1)

{

printf("Press 1 to Add your Loggs\n");

printf("Press 2 to view Your Loggs\n");

printf("Press 3 to Log out\n");

printf("-----");

printf("\n");

scanf("%d",&choice);

printf("-----\n");

switch(choice)

{

case 1:

FILE *ob;

ob = fopen("music_log.txt","w");

printf("File Created! Now Enter Your Loggs\n");

printf("-----\n");

fprintf(ob,"Your Added Log\n");

fprintf(ob,"-----\n");

for(i=1;i<=7;i++)

{

printf("Day %d:-",i);

scanf("%d",&min[i]);

fprintf(ob,"Day %d: %d Minutes\n",i,min[i]);

}

fprintf(ob,"-----\n");

```

        printf("-----\n");
    break ;

    case 2:
        printf("-----\n");
        printf("Your Added Log\n");
        printf("-----\n");
        for(i=1;i<=7;i++)
        {
            printf("Your day %d music play time:- %d Minutes\n",i,min[i]);
            printf("\n");
        }
        printf("-----\n");
    break;

    case 3:
        printf("-----");
        printf("\nLogging out! Visit Again\n");
        printf("-----");
        printf("\n");
    break;

    default:
        printf("-----");
        printf("\nError!!!! Please go with 1 or 2 or 3 only\n");
        printf("-----");
        printf("\n");
    break;
}

if(choice==3)
{
    break;
}

}
}

```

Output:

```
E:\06Aug_Kevin_SE\C_Assign  X  +  v

-----
Welcome to Music Listening App
-----
Press 1 to Add your Loggs
Press 2 to view Your Loggs
Press 3 to Log out
-----
1
-----
File Created! Now Enter Your Loggs
-----
Day 1:-23
Day 2:-33
Day 3:-45
Day 4:-11
Day 5:-23
Day 6:-45
Day 7:-43
-----
Press 1 to Add your Loggs
Press 2 to view Your Loggs
Press 3 to Log out
-----
2
-----
Your Added Log
-----
Your day 1 music play time:- 23 Minutes
Your day 2 music play time:- 33 Minutes
Your day 3 music play time:- 45 Minutes
Your day 4 music play time:- 11 Minutes
Your day 5 music play time:- 23 Minutes
Your day 6 music play time:- 45 Minutes
Your day 7 music play time:- 43 Minutes
-----
Press 1 to Add your Loggs
Press 2 to view Your Loggs
Press 3 to Log out
-----
3
-----
Logging out! Visit Again
-----

-----
Process exited after 26.53 seconds with return value 0
Press any key to continue . . . |
```

S14_T01.cpp X T01.exe X T01.cpp X T02.cpp M X

Mini Project > music_log.txt

1

Your Added Log

2

3

Day 1: 23 Minutes

4

Day 2: 33 Minutes

5

Day 3: 45 Minutes

6

Day 4: 11 Minutes

7

Day 5: 23 Minutes

8

Day 6: 45 Minutes

9

Day 7: 43 Minutes

10

11

```

/*    Tak 4
Add a feature to your app that
reads the saved music_log.txt file
and generates a
weekly report showing
the total,
average, and
highest listening minutes in the week.
*/

#include <stdio.h>

int main()
{
    int min[7];
    int i, choice;

    printf("-----\n");
    printf("Welcome to Music Listening App\n");
    printf("-----\n");

    while(1)
    {
        printf("Press 1 to Add your Logs\n");
        printf("Press 2 to View your Logs (from Memory)\n");
        printf("Press 3 to Generate Weekly Report (from File)\n");
        printf("Press 4 to Log out\n");
        printf("-----\n");
        scanf("%d", &choice);
        printf("-----\n");

        switch(choice)
        {
            case 1: {
                FILE *ob = fopen("music_log.txt", "w");
                if (ob == NULL) {
                    printf("Error opening file for writing!\n");
                    break;
                }

                printf("File Created! Now Enter Your Logs:\n");
                printf("-----\n");
                fprintf(ob, "Your Added Log\n");
                fprintf(ob, "-----\n");
            }
        }
    }
}

```

```

        for(i = 0; i < 7; i++)
        {
            printf("Day %d:- ", i + 1);
            scanf("%d", &min[i]);
            fprintf(ob, "Day %d: %d Minutes\n", i + 1,
min[i]);
        }
        fprintf(ob, "-----\n");
        fclose(ob); // Always close the file stream after writing

        printf("Logs saved successfully to music_log.txt!\n");
        printf("-----\n");
        break;
    }

    case 2: {
        printf("Your Added Log (Current Session)\n");
        printf("-----\n");
        for(i = 0; i < 7; i++)
        {
            printf("Your day %d music play time:- %d Minutes\n", i + 1,
min[i]);
        }
        printf("-----\n");
        break;
    }

    case 3: {
        FILE *ob = fopen("music_log.txt", "r");
        if (ob == NULL) {
            printf("No saved log found! Please add your logs first
(Option 1).\n");
            printf("-----\n");
            break;
        }

        int dayNum, dayMin;
        int totalMinutes = 0;
        int maxMinutes = 0;
        int count = 0;
        char line[256];

        // this will skip header line
        fgets(line, sizeof(line), ob);
        fgets(line, sizeof(line), ob);

```

```

// Read day data from music_log.txt
while (fscanf(ob, "Day %d: %d Minutes\n", &dayNum, &dayMin) == 2)
{
    totalMinutes += dayMin;
    if (dayMin > maxMinutes) {
        maxMinutes = dayMin;
    }
    count++;
}
fclose(ob);

if (count == 0) {
    printf("File is empty or formatted incorrectly.\n");
} else {
    float averageMinutes = (float)totalMinutes / count;

    printf("-----\n");
    printf("          WEEKLY MUSIC REPORT          \n");
    printf("-----\n");
    printf("Total Listening Time    : %d Minutes\n",
totalMinutes);
    printf("Average Daily Listening: %.2f Minutes\n",
averageMinutes);
    printf("Highest Daily Listening: %d Minutes\n", maxMinutes);
    printf("-----\n");
}
break;
}

case 4: {
    printf("Logging out! Visit Again\n");
    printf("-----\n");
    break;
}

default: {
    printf("Error! Please enter 1, 2, 3, or 4 only.\n");
    printf("-----\n");
    break;
}
}

if (choice == 4)
{

```

```

        break;
    }
}
return 0;
}

```

Output:

```

PS E:\06Aug_Kevin_SE\C _ Assignment _ Code\Mini Project> & "E:\06Aug_Kevin_SE\C _ Assignment _ Code\Mini Project\main.exe"
Welcome to Music Listening App
-----
Press 1 to Add your Logs
Press 2 to View your Logs (from Memory)
Press 3 to Generate Weekly Report (from File)
Press 4 to Log out
-----
1
-----
File Created! Now Enter Your Logs:
-----
Day 1:- 30
Day 2:- 40
Day 3:- 20
Day 4:- 15
Day 5:- 60
Day 6:- 33
Day 7:- 24
Logs saved successfully to music_log.txt!
-----
Press 1 to Add your Logs
Press 2 to View your Logs (from Memory)
Press 3 to Generate Weekly Report (from File)
Press 4 to Log out
-----
2
-----
Your Added Log (Current Session)
-----
Your day 1 music play time:- 30 Minutes
Your day 2 music play time:- 40 Minutes
Your day 3 music play time:- 20 Minutes
Your day 4 music play time:- 15 Minutes
Your day 5 music play time:- 60 Minutes
Your day 6 music play time:- 33 Minutes
Your day 7 music play time:- 24 Minutes
-----

```

```

PS E:\06Aug_Kevin_SE\C _ Assignment _ Code\Mini Project> & "E:\06Aug_Kevin_SE\C _ Assignment _ Code\Mini Project\main.exe"
Press 4 to Log out
-----
2
-----
Your Added Log (Current Session)
-----
Your day 1 music play time:- 30 Minutes
Your day 2 music play time:- 40 Minutes
Your day 3 music play time:- 20 Minutes
Your day 4 music play time:- 15 Minutes
Your day 5 music play time:- 60 Minutes
Your day 6 music play time:- 33 Minutes
Your day 7 music play time:- 24 Minutes
-----
Press 1 to Add your Logs
Press 2 to View your Logs (from Memory)
Press 3 to Generate Weekly Report (from File)
Press 4 to Log out
-----
3
-----
WEEKLY MUSIC REPORT
-----
Total Listening Time : 222 Minutes
Average Daily Listening: 31.71 Minutes
Highest Daily Listening: 60 Minutes
-----
Press 1 to Add your Logs
Press 2 to View your Logs (from Memory)
Press 3 to Generate Weekly Report (from File)
Press 4 to Log out
-----
4
-----
Logging out! Visit Again
-----
PS E:\06Aug_Kevin_SE\C _ Assignment _ Code\Mini Project>

```

```

/*
    Refactor your Music Listening Logger
    to allow users to reset their weekly data
    by clearing the array and deleting the contents
    of music_log.txt.

    Constraint:
    Make sure the reset option is available in the
    menu and asks for confirmation before deleting data.
*/

#include <stdio.h>

int main()
{
    int min[7] ;
    int i, choice;

    printf("-----\n");
    printf("Welcome to Music Listening App\n");
    printf("-----\n");

    while(1)
    {
        printf("Press 1 to Add your Logs\n");
        printf("Press 2 to View your Logs (from Memory)\n");
        printf("Press 3 to Generate Weekly Report (from File)\n");
        printf("Press 4 to Reset Weekly Data\n");
        printf("Press 5 to Log out\n");
        printf("-----\n");
        scanf("%d", &choice);
        printf("-----\n");

        switch(choice)
        {
            case 1: {
                FILE *ob = fopen("music_log.txt", "w");
                if (ob == NULL)
                {
                    printf("Error opening file for writing!\n");
                    break;
                }

                printf("File Created! Now Enter Your Logs:\n");
                printf("-----\n");
            }
        }
    }
}

```

```

fprintf(ob, "Your Added Log\n");
fprintf(ob, "-----\n");

for(i = 0; i < 7; i++)
{
    printf("Day %d:- ", i + 1);
    scanf("%d", &min[i]);
    fprintf(ob, "Day %d: %d Minutes\n", i + 1,
min[i]);
}
fprintf(ob, "-----\n");
fclose(ob);

printf("Logs saved successfully to music_log.txt!\n");
printf("-----\n");
break;
}

case 2: {
    printf("Your Added Log (Current Session)\n");
    printf("-----\n");
    for(i = 0; i < 7; i++)
    {
        printf("Your day %d music play time:- %d Minutes\n", i + 1,
min[i]);
    }
    printf("-----\n");
    break;
}

case 3: {
    FILE *ob = fopen("music_log.txt", "r");
    if (ob == NULL)
    {
        printf("No saved log found! Please add your logs first
(option 1).\n");
        printf("-----\n");
        break;
    }

    int dayNum, dayMin;
    int totalMinutes = 0;
    int maxMinutes = 0;
    int count = 0;
    char line[256];

```

```

// this will Skip header lines
fgets(line, sizeof(line), ob);
fgets(line, sizeof(line), ob);

// Reading day data from music_log.txt
while (fscanf(ob, "Day %d: %d Minutes\n", &dayNum, &dayMin) == 2)
{
    totalMinutes += dayMin;
    if (dayMin > maxMinutes) {
        maxMinutes = dayMin;
    }
    count++;
}
fclose(ob);

if (count == 0)
{
    printf("File is empty or formatted incorrectly.\n");
} else
{
    float averageMinutes = (float)totalMinutes / count;

    printf("-----\n");
    printf("          WEEKLY MUSIC REPORT          \n");
    printf("-----\n");
    printf("Total Listening Time   : %d Minutes\n",
totalMinutes);
    printf("Average Daily Listening: %.2f Minutes\n",
averageMinutes);
    printf("Highest Daily Listening: %d Minutes\n", maxMinutes);
    printf("-----\n");
}
break;
}

case 4: {
    char confirm;
    printf("Are you sure you want to reset all weekly data? (y/n):
");
    scanf(" %c", &confirm);

    if (confirm == 'y' || confirm == 'Y') {
        // 1. Reset array elements in memory to 0
        for(i = 0; i < 7; i++)

```

```

        {
            min[i] = 0;
        }

        // 2. Clear file contents by opening in write mode and
closing immediately
        FILE *ob = fopen("music_log.txt", "w");
        if (ob != NULL)
        {
            fclose(ob);
            printf("Weekly data cleared and music_log.txt
deleted/emptied successfully!\n");
        } else
        {
            printf("Error resetting file.\n");
        }
    } else
    {
        printf("Reset operation cancelled.\n");
    }
    printf("-----\n");
    break;
}

case 5:
{
    printf("Logging out! Visit Again\n");
    printf("-----\n");
    break;
}

default:
{
    printf("Error! Please enter 1, 2, 3, 4, or 5 only.\n");
    printf("-----\n");
    break;
}
}

if (choice == 5)
{
    break;
}
}
return 0;

```

}

Output:;



```
File Edit Selection View Go Run Terminal Help Mini Project
Problems 3 Output Terminal Ports Spell Checker 3
• PS E:\06Aug_Kevin_SE\C _ Assignment _ Code\Mini Project> & "E:\06Aug_Kevin_SE\C _ Assignment _ Code\Mini Project\Mini Project.exe"
Welcome to Music Listening App
-----
Press 1 to Add your Logs
Press 2 to View your Logs (from Memory)
Press 3 to Generate Weekly Report (from File)
Press 4 to Reset Weekly Data
Press 5 to Log out
-----
4
-----
Are you sure you want to reset all weekly data? (y/n): y
Weekly data cleared and music_log.txt deleted/emptied successfully!
-----
Press 1 to Add your Logs
Press 2 to View your Logs (from Memory)
Press 3 to Generate Weekly Report (from File)
Press 4 to Reset Weekly Data
Press 5 to Log out
-----
5
-----
Logging out! Visit Again
-----
○ PS E:\06Aug_Kevin_SE\C _ Assignment _ Code\Mini Project> |
```